

KRITHICK JEGATHEESAN

AI Backend Engineer (Python / Golang)

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EXECUTIVE SUMMARY

AI Backend Engineer with 3+ years of hands-on experience in Python, Golang, FastAPI, LangGraph, AWS, Kubernetes, and scalable microservices. Strong experience building AI agent backends, workflow automation systems, async APIs, event-driven architectures, RAG pipelines, and cloud-native platforms with a focus on reliability, scalability, and high-throughput processing.

PROFESSIONAL EXPERIENCE

Full Stack Engineer | [StepsAI](#)

Feb 2025 – Present

- **AI Agent Backend Development:** Built and maintained Python-based AI backend services using **FastAPI**, **Uvicorn**, **Pydantic**, and async I/O for customer agent, copilot, channel, voice, product summary, and product FAQ APIs.
- **Agent Orchestration:** Developed AI agent workflows using **LangGraph** and **LangChain**, including ReAct-style tool-calling loops, streaming responses, message trimming, checkpointing, escalation checks, and conversation state persistence.
- **RAG & Knowledge Retrieval:** Implemented retrieval pipelines using OpenAI embeddings, **Milvus/Zilliz** dense and sparse search, scoped filtering, chunk deduplication, and optional **Cohere** reranking for knowledgebase-backed agent responses.
- **Workflow Automation Platform:** Contributed to a separate workflow service for workflow definitions, triggers, runs, and DAG execution using FastAPI, PostgreSQL, Redis queues, background workers, retry policies, conditional steps, waits, HTTP actions, and connector actions.
- **Integrations & Tools:** Built and supported dynamic tool/provider integrations across web search, knowledgebase, ecommerce, CRM/helpdesk, Gmail, Google Drive, Google Calendar, Outlook, Notion, Meta Integrations(Whatsapp, Instagram, and Messenger) Airtable, Slack, logistics, memory, custom skills, and human handoff.
- **Infrastructure & Observability:** Worked with **PostgreSQL**, **Redis**, **Docker**, **GitHub Actions**, **Sentry**, **OpenTelemetry**, **Langfuse**, JSON logging, token usage tracking, OAuth credential handling, and secure token refresh flows.
- **Team Leadership:** Managed and delegated tasks to 2 interns and 3 engineers, translating customer demo feedback into production-ready AI backend and automation features within 2–3 week sprints.

Backend Developer | [CodingJr](#)

Sep 2024 – Dec 2024

- **Golang API Development:** Built scalable RESTful APIs for a live student ERP platform using Golang and GoFiber.
- **Database Management:** Designed and managed MySQL schemas for high-concurrency workloads.
- **Profiling & Optimization:** Used pprof to optimize CPU and memory bottlenecks, improving performance and efficiency.

Backend Developer | [Virtusa](#)

Feb 2023 – Dec 2023

- **Core Development:** Contributed to real-time applications using Golang and GoFiber.
- **Enterprise Tech:** Worked with Java Spring Boot and applied OOP concepts in enterprise systems.

TECHNICAL SKILLS

Languages: Python, Golang, SQL, TypeScript, Bash, Java

Backend & APIs: FastAPI, Uvicorn, Pydantic, GoFiber, REST APIs, SSE Streaming, Microservices, Async I/O

AI/LLM Engineering: LangGraph, LangChain, OpenAI SDK, Cohere Rerank, Tavily, Mem0, RAG, Tool Calling, Agent Orchestration

Data & Retrieval: PostgreSQL, MySQL, MongoDB, Redis, Milvus, OpenAI Embeddings, Dense/Sparse Search, Kafka

Cloud & DevOps: AWS, Kubernetes (GKE/EKS), Docker, Helm, ArgoCD, GitOps, GitHub Actions, HPA, uv

Observability & Reliability: OpenTelemetry, Sentry, Langfuse, ELK Stack, JSON Logging, Metrics & Alerting, pprof

LIVE PROJECTS

BB-App – Super App | *Golang, Kubernetes, gRPC, MySQL*

[Website](#) | [Play Store](#) | [App Store](#)

- Designed microservices on GCP Kubernetes using **GKE Autopilot** for fully managed scaling and infrastructure.
- Built a gRPC-based database access layer to optimize inter-service communication and reduce latency.
- Leveraged Autopilot for automatic scaling, cost optimization, and reduced operational overhead.
- Implemented Traefik ingress for routing, SSL termination, and service discovery.
- Automated CI/CD using GitHub Actions and container registry workflows.
- Enabled zero-downtime deployments with rolling updates and traffic management.
- Source code is private and not open sourced due to NDA restrictions.

UI-Network – Multi-Level Marketing Application | *Golang, PostgreSQL, OpenTelemetry*

[Website](#)

- Designed multi-tenant relational schemas with strong isolation guarantees.
- Optimized high-write transactional queries for scalability and low latency.
- Implemented distributed tracing using OpenTelemetry with ELK integration.
- Deployed and managed services on **AWS EKS** with Kubernetes and leveraged **HPA** for auto-scaling.
- Integrated AWS S3 using pre-signed URLs for secure file handling.
- Built fault-tolerant infrastructure with rolling deployments and scaling.

Elderly – NRI Support Mobile Application | *Backend Developer*

[Play Store](#)

- Worked on backend service development, API integrations, database operations, and performance-focused improvements for mobile app workflows.

EDUCATION

KCG College of Technology

B.E. in Computer Science Engineering

Chennai, India

Graduated: May 2025